

FoldShield⁺⁺ Structural Analysis Report

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Input Structures

Reference PDB	1TIM.pdb
Query PDB	4TIM.pdb
Mode	ca
Ref chain	A
Query chain	A

Classical Coordinate Metrics

These metrics saturate near 1.0 even when function diverges. Use FoldShield++ signals below for structural perturbation insight.

Metric	Value	Note
TM-score	1.000	>=0.50 same fold · >=0.90 nearly identical
RMSD (A)	1.300	Lower is better · >3.0 A = significant deviation

FoldShield++ Signal Scores — Phase 1 + Phase 2

All active signals with scores, fusion weights, and interpretation. Phase 2 signals are highlighted in blue.

Braid similarity	0.712	0.20	strong braid similarity
Motif similarity (UL-DSL)	0.734	0.37	moderate motif similarity
SSEF entropy (Phase 2) [P2]	0.882	0.20	minor symbolic grammar rewrite — active-site mutation regime
Persistent homology (Phase 3) [P2]	0.862	0.23	similar topological persistence structure

Fusion Layer v2 — Combined Score

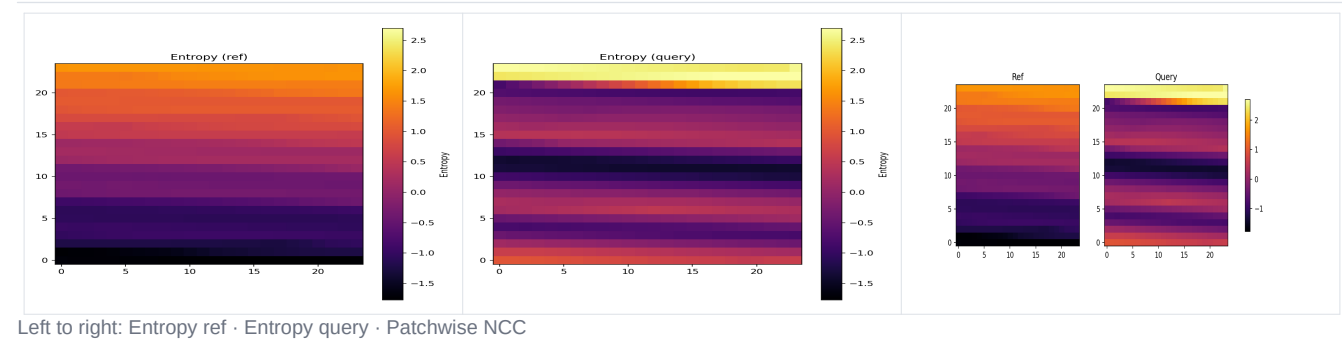
0.7884	phase2	highly similar structural architecture

Phase 2 — Perturbation Classification

SSEF score	Local topology	Gap score	Perturbation class
0.882	0.332	0.550	Conformational transition

Gap = SSEF minus local topology. <0.15 buried-core mutation | 0.15-0.36 fold-switch | >0.36 conformational transition

Entropy Maps & Signal Profiles



Signal Score Bars

Visual 0-1 scale for all active signals. Green ≥ 0.66 / Amber 0.33-0.66 / Red < 0.33 .



Symbolic Score Detail

Symbolic Similarity Score	0.712	overall nearly identical
Behavioural Correlation Similarity	0.546	moderate entropy similarity
Structural Topology Similarity	0.996	identical set of generators
Structural Topology Ordering Similarity	0.702	strong pattern similarity

Signal Analysis — Figures

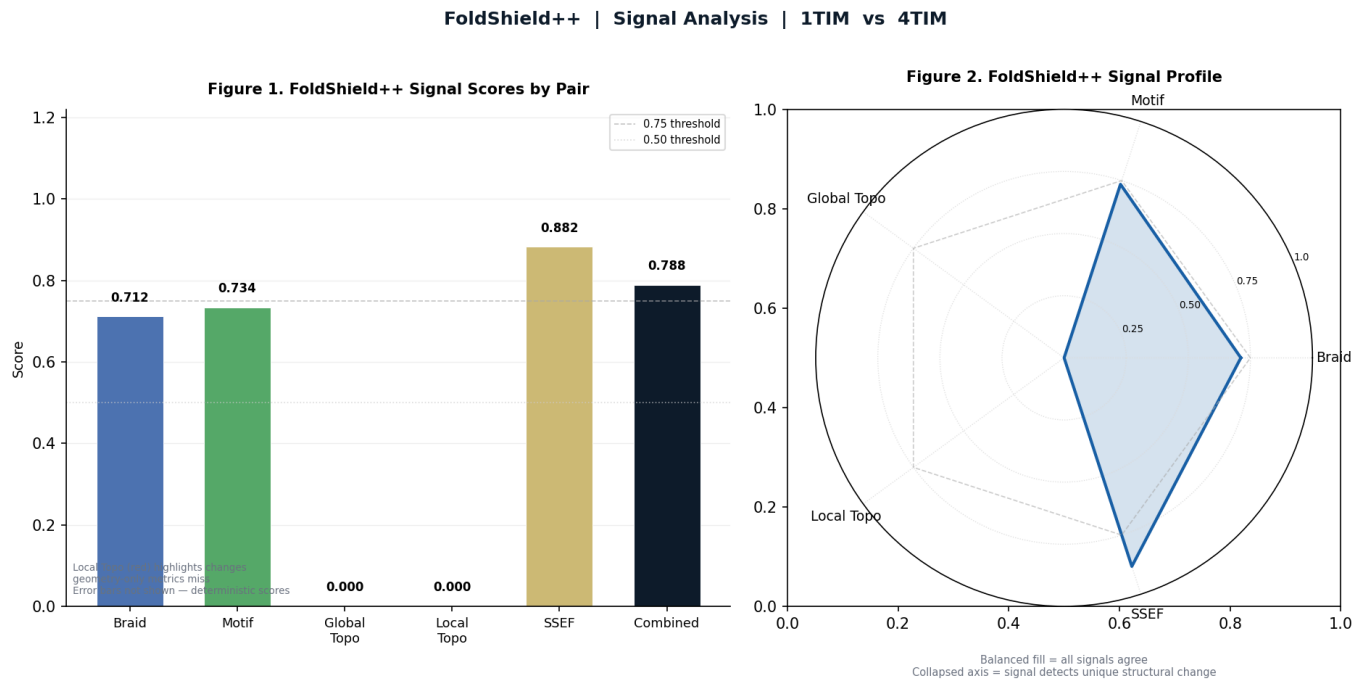


Figure 1: Bar chart comparing all five signals and the Combined score. Local Topo (red) highlights structural changes geometry-only metrics miss. Dashed = 0.75 · Dotted = 0.50. Error bars not shown — scores are deterministic. Figure 2: Radar chart showing the five-signal profile. Balanced fill = all signals agree. Collapsed axis = signal detects unique structural change.